



SYNTHESIS OF COF FOR POST-COMBUSTION UTILISATION FOR CO2

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ABOUT CO2

Carbon dioxide (CO₂) is a major greenhouse gas emitted from power plants, industries, and combustion processes. Its accumulation in the atmosphere drives global warming, making efficient CO₂ capture essential for climate mitigation and industrial decarbonisation.

- 1.CO₂ is a major greenhouse gas with higher density and moderate solubility, making it easier to separate from industrial flue gases.
- 2.Its acidic nature and strong interaction with nitrogen-rich or basic sites enable highly selective adsorption during CO₂ capture.

Objectives of the Project :

1. Capture CO₂ from Industrial Flue Gas

To design and optimize materials (COFs) capable of selectively separating CO₂ from post-combustion gas streams containing N₂, O₂, SOx, and moisture.

2. Improve Efficiency & Reduce Energy Costs

To develop a low-energy, high-selectivity capture system that outperforms traditional MEA-based methods in regeneration energy, corrosion, and operational cost.

3. Use Environment-Friendly, Scalable Materials

To synthesize TAPB-PDA COF using a green solvent system (n-butanol-acetic acid-water), ensuring sustainability, safety, and industrial scalability.

4. Enhance CO₂ Adsorption & Selectivity

To functionalize and optimize COF pores to achieve higher CO₂ uptake, better CO₂/N₂ separation, and increased recycling stability.

5. Enable CO₂ Utilisation (CCU)

To convert captured CO₂ into valuable products such as:
Methanol
Urea
Polymers
Carbonates (mineralisation)
Feedstock for enhanced oil recovery (EOR)

TAPB – PDA + n-butanol acetic Acid Mixture

Procedure and why we chose this compound :

Composition & Material to be used for this reaction :

1,3,5-Tris(4-aminophenyl)benzene (TAPB) and Terephthalaldehyde (Ta)
Solvent System: n-Butanol (n-BuOH), Deionized Water (H₂O), and Glacial Acetic Acid (AcOH)
Washing Solvents: Methanol, Acetone, and Diethyl Ether

n-Butanol	0.503	0.81 (typical at 20–25 °C).	$0.503 \times 0.81 = \mathbf{0.407\text{ g}}$
Water (deionized)	0.075	1.000 (at 20–25 °C)	$0.075 \times 1.000 = \mathbf{0.075\text{ g}}$
Glacial acetic acid	0.172	1.049 (at 25 °C)	$0.172 \times 1.049 = \mathbf{0.180\text{ g}}$

Traditionally, TAPB-PDA COF is synthesized in a 1,4-dioxane-mesitylene mixture at 110 °C for several days, giving high porosity but only moderate yields and safety concerns. The n-butanol-acetic acid-water system delivers comparable crystallinity and BET surface area under milder, greener conditions (70 °C, 16 h). Its optimized solvent balance improves reversible imine bond formation, boosting uniformity and raising yields to 90–95%. This method also lowers energy use, enhances scalability, and improves sustainability—making n-butanol the preferred solvent for high-yield, industrial CO₂-capture-grade TAPB-PDA COF production.

Procedure and Post Characterisation :

1 Weigh and Combine Monomers

Weigh 10.5 mg TAPB and 6.0 mg PDA into a clean 4 mL vial, maintaining the correct 1:1.5 molar ratio. Add a 1 cm Teflon stir bar to ensure proper mixing during the reaction.

4 Run the Reaction

Place the sealed vial in a heating block at 70 °C and stir at 250 RPM for 16 hours. The reaction proceeds under ambient air, requiring no inert atmosphere.

2 Prepare the Solvent Mixture

Prepare 0.75 mL of solvent by mixing n-butanol, water, and acetic acid in the order listed. This specific ratio and sequence ensure the correct acidity and solubility for COF formation.

5 Cool, Filter, and Wash

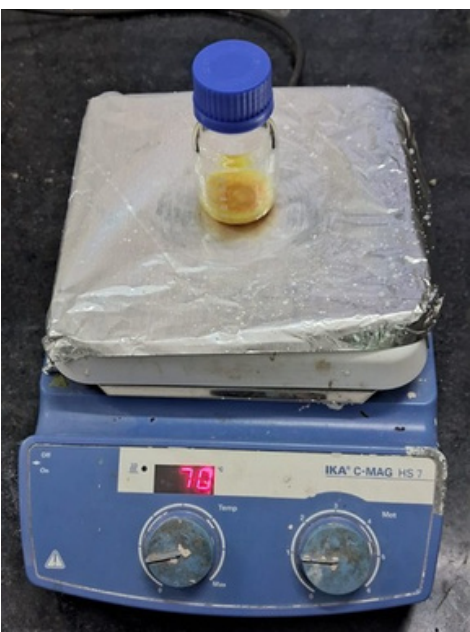
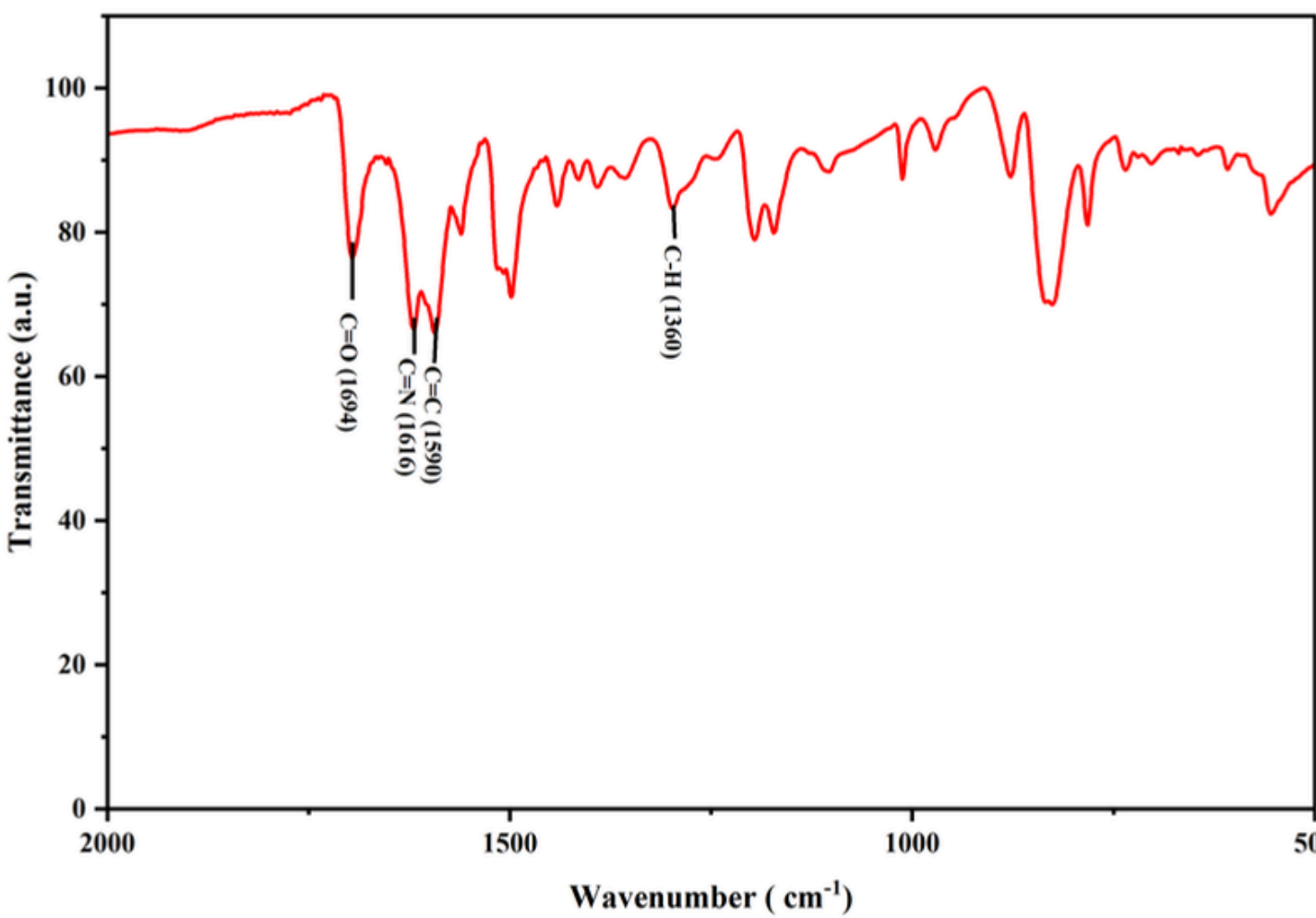
Allow the vial to cool, then filter the suspension through a 0.45 µm nylon filter. Wash the solid sequentially with water, methanol (twice), and acetone (twice), keeping the material wet at all times.

3 Add Solvent and Seal the Vial

Add the prepared solvent mixture to the monomers and gently swirl to wet the solids. Seal the vial tightly with a PTFE-lined cap to prevent evaporation during heating.

6 Add Solvent and Seal the Vial

Activate the wet COF by washing and soaking it in diethyl ether, then air-drying for 16 hours. Alternatively, dry it in a vacuum oven at 90 °C for 16 hours.



Results :

We synthesized the TAPB-Ta COF (using TAPB and Terephthalaldehyde). The experiment successfully synthesized COF, confirmed by the formation of the imine linkage (C=N) peak at 1616 cm⁻¹ in the FTIR spectrum. However, the spectrum also shows a strong residual carbonyl (C=O) peak at 1694 cm⁻¹, indicating that unreacted terephthalaldehyde is currently trapped in the material and requires further solvent washing to purify.

Signature of Supervisor :